



Continuum Mechanics

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Chapter 02

Tensor Calculus

Part 2



Divergence and Product Rules



Definition (2.7)

Let $\Omega \subseteq \mathbb{E}^3$ be open. To every differentiable vector field $\mathbf{v} : \Omega \rightarrow \mathcal{V}$, we associate the scalar field

$$\nabla \cdot \mathbf{v} : \Omega \longrightarrow \mathbb{R}$$

defined by

$$\nabla \cdot \mathbf{v} = \text{tr}(\nabla \mathbf{v}).$$

The scalar field $\nabla \cdot \mathbf{v}$ is called the *divergence* of $\mathbf{v}$.



Result 2.8

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame. Write

$$\mathbf{v}(\mathbf{x}) = v_i(\mathbf{x})\mathbf{e}_i,$$

and assume that $\mathbf{v} : \Omega \rightarrow \mathcal{V}$ is differentiable. Then

$$\nabla \cdot \mathbf{v}(\mathbf{x}) = \frac{\partial v_i}{\partial x_i}(\mathbf{x}) = v_{i,i}(\mathbf{x}).$$

The repeated index i is a dummy index.

Proof.

Exercise.





Definition (2.9)

Let $\Omega \subseteq \mathbb{E}^3$ be open. To every differentiable second-order tensor field

$$S : \Omega \longrightarrow \mathcal{V}^2,$$

we associate the vector field

$$\nabla \cdot S : \Omega \longrightarrow \mathcal{V}$$

defined by the relation

$$(\nabla \cdot S) \cdot \mathbf{a} = \nabla \cdot (S^T \mathbf{a})$$

for every constant vector $\mathbf{a} \in \mathcal{V}$.

We call $\nabla \cdot S$ the *divergence* of S .



Result 2.10

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame and assume that $S : \Omega \rightarrow \mathcal{V}^2$ is differentiable. If

$$S = S_{ij}\mathbf{e}_i \otimes \mathbf{e}_j,$$

then

$$(\nabla \cdot S)(\mathbf{x}) = \frac{\partial S_{ij}}{\partial x_j}(\mathbf{x})\mathbf{e}_i = S_{ij,j}(\mathbf{x})\mathbf{e}_i.$$

Equivalently,

$$[\nabla \cdot S]_i = S_{ij,j}.$$

In this component identity i is free and j is dummy; in the vector expansion above, both indices are summed.



Proof.

Let $\mathbf{a} = a_k \mathbf{e}_k \in \mathcal{V}$ be an arbitrary constant vector and set

$$\mathbf{q} := S^T \mathbf{a}.$$

Write $\mathbf{q} = q_j \mathbf{e}_j$. Since

$$S = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j,$$

we have

$$q_j = [\mathbf{q}]_j = S_{ij} a_i.$$

Using Definition 2.9 and Result 2.8,

$$\begin{aligned} (\nabla \cdot S) \cdot \mathbf{a} &= \nabla \cdot (S^T \mathbf{a}) \\ &= \nabla \cdot \mathbf{q} \\ &= q_{j,j}. \end{aligned}$$



Proof (Cont.)

Because $\mathbf{a}$ is constant,

$$\begin{aligned}
 (\nabla \cdot S) \cdot \mathbf{a} &= \frac{\partial q_j}{\partial x_j} \\
 &= \frac{\partial}{\partial x_j} (S_{ij} a_i) \\
 &= \frac{\partial S_{ij}}{\partial x_j} a_i \\
 &= S_{ij,j} a_i \\
 &= (S_{ij,j} \mathbf{e}_i) \cdot \mathbf{a}.
 \end{aligned}$$

Since $\mathbf{a} \in \mathcal{V}$ is arbitrary,

$$\nabla \cdot S = S_{ij,j} \mathbf{e}_i.$$





Result 2.11 (Product Rules)

Let ϕ , $\mathbf{v}$, and S be differentiable scalar, vector, and tensor fields, respectively. Then

$$\nabla(\phi\mathbf{v}) = \mathbf{v} \otimes \nabla\phi + \phi\nabla\mathbf{v},$$

$$\nabla \cdot (\phi S) = \phi \nabla \cdot S + S \nabla \phi,$$

$$\nabla \cdot (S^T \mathbf{v}) = (\nabla \cdot S) \cdot \mathbf{v} + S : \nabla \mathbf{v}.$$



Proof.

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame and write

$$\mathbf{v} = v_i \mathbf{e}_i, \quad S = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Using Result 2.6,

$$\begin{aligned} \nabla(\phi \mathbf{v}) &= \frac{\partial(\phi v_i)}{\partial x_j} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= \frac{\partial \phi}{\partial x_j} v_i \mathbf{e}_i \otimes \mathbf{e}_j + \phi \frac{\partial v_i}{\partial x_j} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= (v_i \mathbf{e}_i) \otimes \left(\frac{\partial \phi}{\partial x_j} \mathbf{e}_j \right) + \phi \nabla \mathbf{v} \\ &= \mathbf{v} \otimes \nabla \phi + \phi \nabla \mathbf{v}. \end{aligned}$$



Proof (Cont.)

Next, using Result 2.10,

$$\begin{aligned}\nabla \cdot (\phi S) &= \frac{\partial(\phi S_{ij})}{\partial x_j} \mathbf{e}_i \\ &= \left(\frac{\partial \phi}{\partial x_j} S_{ij} + \phi \frac{\partial S_{ij}}{\partial x_j} \right) \mathbf{e}_i \\ &= S_{ij} \frac{\partial \phi}{\partial x_j} \mathbf{e}_i + \phi \nabla \cdot S \\ &= S \nabla \phi + \phi \nabla \cdot S.\end{aligned}$$



Proof (Cont.)

Finally, $S^T \mathbf{v}$ is a vector field. In the frame $\mathcal{F}$,

$$\begin{aligned} S^T \mathbf{v} &= [S^T]_{ij} v_j \mathbf{e}_i \\ &= S_{ji} v_j \mathbf{e}_i \\ &= S_{ij} v_i \mathbf{e}_j, \end{aligned}$$

where the last equality consistently renames the dummy indices.

Using Result 2.8,

$$\begin{aligned} \nabla \cdot (S^T \mathbf{v}) &= \frac{\partial (S_{ij} v_i)}{\partial x_j} \\ &= \frac{\partial S_{ij}}{\partial x_j} v_i + S_{ij} \frac{\partial v_i}{\partial x_j}. \end{aligned}$$



Proof (Cont.)

The first term is

$$\frac{\partial S_{ij}}{\partial x_j} v_i = (\nabla \cdot S) \cdot \mathbf{v}.$$

The second term is

$$S_{ij} \frac{\partial v_i}{\partial x_j} = S_{ij} [\nabla \mathbf{v}]_{ij} = S : \nabla \mathbf{v}.$$

Therefore,

$$\nabla \cdot (S^T \mathbf{v}) = (\nabla \cdot S) \cdot \mathbf{v} + S : \nabla \mathbf{v}.$$





Worked Exercises



Example (Exercise 2.4)

Let $\Omega \subseteq \mathbb{R}^3$ be open, and let

$$\phi : \Omega \rightarrow \mathbb{R}, \quad \mathbf{v}, \mathbf{w} : \Omega \rightarrow \mathcal{V}, \quad S : \Omega \rightarrow \mathcal{V}^2$$

be differentiable fields. Show that

(a)

$$\nabla \cdot (\phi \mathbf{v}) = \nabla \phi \cdot \mathbf{v} + \phi \nabla \cdot \mathbf{v};$$

(b)

$$\nabla(\mathbf{v} \cdot \mathbf{w}) = (\nabla \mathbf{v})^T \mathbf{w} + (\nabla \mathbf{w})^T \mathbf{v};$$

(c)

$$\nabla \cdot (\mathbf{v} \otimes \mathbf{w}) = (\nabla \mathbf{v}) \mathbf{w} + (\nabla \cdot \mathbf{w}) \mathbf{v}.$$



Example (Exercise 2.4(a))

Using Result 2.8 and the scalar product rule,

$$\begin{aligned}\nabla \cdot (\phi \mathbf{v}) &= \frac{\partial}{\partial x_i} (\phi v_i) \\ &= \frac{\partial \phi}{\partial x_i} v_i + \phi \frac{\partial v_i}{\partial x_i} \\ &= \nabla \phi \cdot \mathbf{v} + \phi (\nabla \cdot \mathbf{v}).\end{aligned}$$

Here i is a dummy index in every term; there is no free index because the result is a scalar.



Example (Exercise 2.4(b))

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame. Since

$$\mathbf{v} \cdot \mathbf{w} = v_k w_k,$$

we have

$$\begin{aligned} \nabla(\mathbf{v} \cdot \mathbf{w}) &= \frac{\partial}{\partial x_i} (v_k w_k) \mathbf{e}_i \\ &= \left(\frac{\partial v_k}{\partial x_i} w_k + v_k \frac{\partial w_k}{\partial x_i} \right) \mathbf{e}_i \\ &= \left([(\nabla \mathbf{v})^T]_{ik} w_k + [(\nabla \mathbf{w})^T]_{ik} v_k \right) \mathbf{e}_i. \end{aligned}$$

Here i is free in the component identity and k is dummy.



Example (Exercise 2.4(b), Cont.)

Using square brackets for component extraction,

$$\left[(\nabla \mathbf{v})^T \mathbf{w} \right]_i = [(\nabla \mathbf{v})^T]_{ik} w_k,$$

$$\left[(\nabla \mathbf{w})^T \mathbf{v} \right]_i = [(\nabla \mathbf{w})^T]_{ik} v_k.$$

Therefore,

$$\nabla(\mathbf{v} \cdot \mathbf{w}) = (\nabla \mathbf{v})^T \mathbf{w} + (\nabla \mathbf{w})^T \mathbf{v}.$$

Both sides have the same free component index i ; k is dummy.



Example (Exercise 2.4(c))

The components of the dyad $\mathbf{v} \otimes \mathbf{w}$ are

$$[\mathbf{v} \otimes \mathbf{w}]_{ij} = v_i w_j.$$

Thus, by Result 2.10,

$$\begin{aligned} \nabla \cdot (\mathbf{v} \otimes \mathbf{w}) &= \frac{\partial}{\partial x_j} (v_i w_j) \mathbf{e}_i \\ &= \left(\frac{\partial v_i}{\partial x_j} w_j + v_i \frac{\partial w_j}{\partial x_j} \right) \mathbf{e}_i. \end{aligned}$$

Here i is free in the component identity and j is dummy.



Example (Exercise 2.4(c), Cont.)

The first term satisfies

$$\frac{\partial v_i}{\partial x_j} w_j = [(\nabla \mathbf{v}) \mathbf{w}]_i,$$

and the second satisfies

$$v_i \frac{\partial w_j}{\partial x_j} = [(\nabla \cdot \mathbf{w}) \mathbf{v}]_i.$$

Therefore,

$$\nabla \cdot (\mathbf{v} \otimes \mathbf{w}) = (\nabla \mathbf{v}) \mathbf{w} + (\nabla \cdot \mathbf{w}) \mathbf{v}.$$

Again, i is free and j is dummy in the component calculation.



The Curl Operator



Definition (2.12)

To every differentiable vector field

$$\mathbf{v} : \mathbb{E}^3 \longrightarrow \mathcal{V},$$

we associate another vector field

$$\nabla \times \mathbf{v} : \mathbb{E}^3 \longrightarrow \mathcal{V}$$

defined by

$$(\nabla \times \mathbf{v}) \times \mathbf{a} = (\nabla \mathbf{v} - (\nabla \mathbf{v})^T) \mathbf{a}$$

for every constant vector $\mathbf{a} \in \mathcal{V}$. We call $\nabla \times \mathbf{v}$ the *curl* of $\mathbf{v}$.



Remark

At a point $\mathbf{x}$, if $\mathbf{v}$ is a velocity field, then

$$(\nabla \times \mathbf{v})(\mathbf{x})$$

describes the rate and direction of the rotation at $\mathbf{x}$.

The definition uses the cross product through its linearity and its action on frame vectors. In a right-handed orthonormal frame,

$$\mathbf{e}_i \times \mathbf{e}_j = \epsilon_{ijk} \mathbf{e}_k.$$

The permutation symbol ϵ_{ijk} was defined in Class 01.



Result 2.13

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be a right-handed orthonormal frame and let

$$\mathbf{v}(\mathbf{x}) = v_i(\mathbf{x})\mathbf{e}_i.$$

Then

$$(\nabla \times \mathbf{v})(\mathbf{x}) = \epsilon_{ijk} v_{i,k}(\mathbf{x})\mathbf{e}_j,$$

where

$$v_{i,k} = \frac{\partial v_i}{\partial x_k}.$$

All three indices i, j, k are summed in the vector expansion.



Result 2.13 (Cont.)

Equivalently,

$$\begin{aligned}\nabla \times \mathbf{v} &= \left(\frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3} \right) \mathbf{e}_1 \\ &\quad + \left(\frac{\partial v_1}{\partial x_3} - \frac{\partial v_3}{\partial x_1} \right) \mathbf{e}_2 \\ &\quad + \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) \mathbf{e}_3.\end{aligned}$$



Proof.

Recall Result 1.3. If W is skew-symmetric, there is a unique axial vector $\mathbf{w}$ such that

$$\mathbf{w} \times \mathbf{a} = W\mathbf{a}, \quad \mathbf{a} \in \mathcal{V}.$$

Since

$$\nabla \mathbf{v} - (\nabla \mathbf{v})^T$$

is skew-symmetric, there is a unique vector $\nabla \times \mathbf{v}$ such that

$$(\nabla \times \mathbf{v}) \times \mathbf{a} = (\nabla \mathbf{v} - (\nabla \mathbf{v})^T)\mathbf{a}, \quad \mathbf{a} \in \mathcal{V}.$$

Thus the curl is well-defined.



Proof (Cont.)

Set

$$\mathbf{w} = \nabla \times \mathbf{v}, \quad W = \nabla \mathbf{v} - (\nabla \mathbf{v})^T.$$

By Result 1.3,

$$\begin{aligned} w_j = [\mathbf{w}]_j &= \frac{1}{2} \epsilon_{ijk} W_{ik} \\ &= \frac{1}{2} \epsilon_{ijk} (v_{i,k} - v_{k,i}) \\ &= \frac{1}{2} \epsilon_{ijk} v_{i,k} - \frac{1}{2} \epsilon_{ijk} v_{k,i}. \end{aligned}$$

Here j is free, while i, k are dummy indices.



Proof (Cont.)

In the second term, consistently interchange the dummy indices i and k :

$$\begin{aligned}
 w_j &= \frac{1}{2} \epsilon_{ijk} v_{i,k} - \frac{1}{2} \epsilon_{kji} v_{i,k} \\
 &= \frac{1}{2} \epsilon_{ijk} v_{i,k} + \frac{1}{2} \epsilon_{ijk} v_{i,k} \quad (\epsilon_{kji} = -\epsilon_{ijk}) \\
 &= \epsilon_{ijk} v_{i,k}.
 \end{aligned}$$

Hence

$$\nabla \times \mathbf{v} = \mathbf{w} = w_j \mathbf{e}_j = \epsilon_{ijk} v_{i,k} \mathbf{e}_j.$$





The Laplacian Operator



Definition (2.14)

To every twice differentiable scalar field

$$\phi : \mathbb{E}^3 \longrightarrow \mathbb{R},$$

we associate the scalar field

$$\Delta\phi := \nabla \cdot (\nabla\phi).$$

To every twice differentiable vector field

$$\mathbf{v} : \mathbb{E}^3 \longrightarrow \mathcal{V},$$

we associate the vector field

$$\Delta\mathbf{v} := \nabla \cdot (\nabla\mathbf{v}).$$

We call $\Delta\phi$ and $\Delta\mathbf{v}$ the *Laplacians* of ϕ and $\mathbf{v}$, respectively.



Result 2.15

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame and write

$$\mathbf{v}(\mathbf{x}) = v_i(\mathbf{x})\mathbf{e}_i.$$

Then

$$\Delta\phi(\mathbf{x}) = \phi_{,ii}(\mathbf{x})$$

and

$$\Delta\mathbf{v}(\mathbf{x}) = v_{i,jj}(\mathbf{x})\mathbf{e}_i,$$

where (x_1, x_2, x_3) are the coordinates of $\mathbf{x}$ in $\mathcal{F}$.

In the scalar formula i is dummy. In the vector expansion i and j are both summed.



Proof.

By Result 2.4,

$$\nabla\phi = \phi_{,i}\mathbf{e}_i.$$

Using Result 2.8,

$$\begin{aligned}\Delta\phi &= \nabla \cdot (\nabla\phi) \\ &= (\phi_{,i})_{,i} \\ &= \phi_{,ii}.\end{aligned}$$



Proof (Cont.)

By Result 2.6,

$$\nabla \mathbf{v} = v_{i,j} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Using Result 2.10,

$$\begin{aligned} \Delta \mathbf{v} &= \nabla \cdot (\nabla \mathbf{v}) \\ &= (v_{i,j})_{,j} \mathbf{e}_i \\ &= v_{i,jj} \mathbf{e}_i. \end{aligned}$$





The Divergence Theorems

Regular Regions



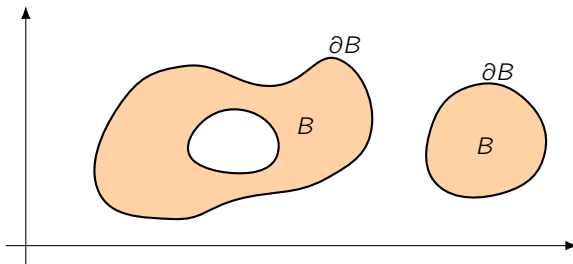
Definition

A region $B \subseteq \mathbb{E}^3$ is called *regular* if and only if

- ❶ B consists of a finite number of open, disjoint, bounded components;
- ❷ the boundary surface ∂B is piecewise smooth and consists of a finite number of disjoint components;
- ❸ each component of ∂B is orientable in the sense that it clearly has two sides.

A regular region can have holes and disconnected components.

Regular Regions



The normal $\hat{\mathbf{n}}$ points outward on every component of ∂B , including the boundary of a hole.



Result 2.16 (Divergence Theorem)

Let B denote a regular region in $\mathbb{E}^3$ with a piecewise smooth boundary ∂B , and consider a differentiable vector field $\mathbf{v} : B \rightarrow \mathcal{V}$. Then

$$\int_{\partial B} \mathbf{v} \cdot \hat{\mathbf{n}} \, dA = \int_B \nabla \cdot \mathbf{v} \, dV,$$

where $\hat{\mathbf{n}}$ is the outward-pointing unit normal field on ∂B .

In component form,

$$\int_{\partial B} v_i n_i \, dA = \int_B v_{i,i} \, dV.$$

Proof.

See M. Spivak, *Calculus on Manifolds*, Addison–Wesley, 1965. □

Flux Across a Surface



The surface integral

$$\int_{\partial B} \mathbf{v} \cdot \hat{\mathbf{n}} \, dA = \int_{\partial B} v_i n_i \, dA$$

is called the *flux* of $\mathbf{v}$ across the surface ∂B .

The divergence theorem equates this outward flux to the volume integral of the divergence:

$$\int_{\partial B} \mathbf{v} \cdot \hat{\mathbf{n}} \, dA = \int_B \nabla \cdot \mathbf{v} \, dV.$$



Result 2.17 (Tensor Divergence Theorem)

Let B denote a regular region in $\mathbb{E}^3$ with piecewise smooth boundary ∂B , and consider a differentiable second-order tensor field

$$S : B \longrightarrow \mathcal{V}^2.$$

Then

$$\int_{\partial B} S \hat{\mathbf{n}} \, dA = \int_B \nabla \cdot S \, dV.$$

In component form,

$$\int_{\partial B} S_{ij} n_j \, dA = \int_B S_{ij,j} \, dV.$$

In the component identity, i is free and j is dummy.



Proof.

Let $\mathbf{a} \in \mathcal{V}$ be any constant vector. Then

$$\begin{aligned}\mathbf{a} \cdot \int_{\partial B} S \hat{\mathbf{n}} \, dA &= \int_{\partial B} \mathbf{a} \cdot (S \hat{\mathbf{n}}) \, dA \\ &= \int_{\partial B} (S^T \mathbf{a}) \cdot \hat{\mathbf{n}} \, dA.\end{aligned}$$

Since $S^T \mathbf{a}$ is a vector field and $\mathbf{a}$ is constant,

$$\begin{aligned}\nabla \cdot (S^T \mathbf{a}) &= \frac{\partial}{\partial x_i} (S_{ji} a_j) \\ &= \frac{\partial S_{ji}}{\partial x_i} a_j \\ &= (\nabla \cdot S) \cdot \mathbf{a}.\end{aligned}$$



Proof (Cont.)

Applying the divergence theorem to the vector field $S^T \mathbf{a}$ gives

$$\begin{aligned}
 \mathbf{a} \cdot \int_{\partial B} S \hat{\mathbf{n}} \, dA &= \int_{\partial B} (S^T \mathbf{a}) \cdot \hat{\mathbf{n}} \, dA \\
 &= \int_B \nabla \cdot (S^T \mathbf{a}) \, dV \\
 &= \int_B (\nabla \cdot S) \cdot \mathbf{a} \, dV \\
 &= \mathbf{a} \cdot \int_B \nabla \cdot S \, dV.
 \end{aligned}$$

Since $\mathbf{a} \in \mathcal{V}$ was arbitrary,

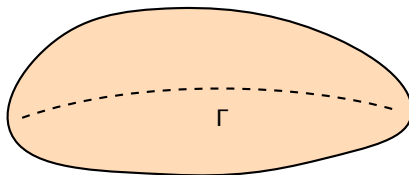
$$\int_{\partial B} S \hat{\mathbf{n}} \, dA = \int_B \nabla \cdot S \, dV.$$





Stokes' Theorem

Orientation of a Surface and Its Boundary



$$C = \partial\Gamma$$

The orientation of the unit tangent $\hat{\mathbf{t}}$ on C is determined by the right-hand rule from the chosen unit normal $\hat{\mathbf{n}}$ on Γ .



Result 2.18 (Stokes' Theorem)

Let Γ be a surface in $\mathbb{E}^3$ with a piecewise smooth boundary curve $C = \partial\Gamma$, and consider a differentiable vector field $\mathbf{v} : \mathbb{E}^3 \rightarrow \mathcal{V}$. Then

$$\int_{\Gamma} (\nabla \times \mathbf{v}) \cdot \hat{\mathbf{n}} \, dA = \int_C \mathbf{v} \cdot \hat{\mathbf{t}} \, ds,$$

where $\hat{\mathbf{n}}$ is a unit normal field on Γ and $\hat{\mathbf{t}}$ is a unit tangent field on C , oriented in the right-handed fashion.

The integral

$$\int_C \mathbf{v} \cdot \hat{\mathbf{t}} \, ds$$

is called the *circulation* of $\mathbf{v}$ around C .

Proof.

See M. Spivak, *Calculus on Manifolds*, Addison–Wesley, 1965. □



The Localization Theorem



Result 2.19 (Localization Theorem)

Consider an open set $B \subseteq \mathbb{E}^3$ and a continuous function

$$\phi : B \longrightarrow \mathbb{R}.$$

Let Ω denote an arbitrary open subset of B . If

$$\int_{\Omega} \phi(\mathbf{x}) \, dV = 0 \quad \text{for every open } \Omega \subseteq B, \quad (2.1)$$

then

$$\phi(\mathbf{x}) = 0 \quad \text{for every } \mathbf{x} \in B. \quad (2.2)$$



Proof.

Assume that (2.1) is true, but (2.2) is false. Then there is some point $\mathbf{y} \in B$ for which

$$\phi(\mathbf{y}) \neq 0.$$

Suppose that

$$\phi(\mathbf{y}) = 2\delta, \quad \delta > 0.$$

The case $\phi(\mathbf{y}) < 0$ can be handled similarly.

Since ϕ is continuous, there is an open neighborhood Ω of $\mathbf{y}$ such that

$$\mathbf{y} \in \Omega \subseteq B$$

and

$$\phi(\mathbf{x}) > \delta, \quad \mathbf{x} \in \Omega.$$



Proof (Cont.)

For this open set Ω ,

$$\int_{\Omega} \phi(\mathbf{x}) \, dV > \delta \, \text{Vol}(\Omega) > 0.$$

This contradicts (2.1), which asserts that

$$\int_{\Omega} \phi(\mathbf{x}) \, dV = 0$$

for every open $\Omega \subseteq B$. Hence (2.2) is true:

$$\phi(\mathbf{x}) = 0, \quad \mathbf{x} \in B.$$

